Assignment 3 — IoT Sensor Data Analysis

Dataset

iot_sensor_data_raw.csv

Objective

You are working as an IoT data analyst for a manufacturing company.

The company has multiple sensors installed across factories. Your task is to analyze the sensor data and identify abnormal operating conditions.

Part A — Data Preparation

1. Load the dataset.
2. Display its dimensions and structure.
3. Convert Timestamp into Pandas datetime.
4. Check whether the timestamps are correctly ordered.
5. Identify missing sensor readings.
6. Calculate the percentage of missing values for each sensor.

Part B — Data Cleaning

7. Handle missing sensor values using an appropriate method.
8. Identify abnormal temperature readings.
9. Identify abnormal vibration readings.
10. Identify machines with critically low battery levels.
11. Explain how you decided what constitutes an abnormal reading.

Part C — Time-Series Analysis

12. Calculate the average temperature by hour.
13. Calculate the average temperature for each device.
14. Calculate the average vibration for each device.
15. Find the maximum temperature recorded by each device.
16. Find the minimum battery level for each device.
17. Determine which device has the highest average vibration.
18. Determine which factory has the highest average temperature.

Part D — Create New Variables

19. Create a Battery_Status column:

Battery	Status
>= 50	Healthy

20–49	Moderate
<20	Critical

20. Create a Temperature_Status column.
21. Create a Vibration_Status column.
22. Create an overall Machine_Health column:

Normal
Warning
Critical

based on appropriate sensor conditions.

Part E — Advanced Analysis

23. Find devices that have experienced critical conditions more than five times.
24. Find the factory with the highest number of abnormal sensor readings.
25. Calculate the percentage of time each device operates under warning/critical conditions.
26. Identify the device that requires the highest maintenance priority.